

Predicting Ideology from Social Media Content Using Weakly Supervised Transformer Models

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Abstract

This project studies whether social media text can support fine-grained political ideology prediction beyond a single left-right label. The system design combines weak supervision, disagreement-aware annotation, flat and hierarchical transformer baselines, calibration, counterfactual checks, and error analysis. The completed real-data experiment uses MITweet with pretrained RoBERTa and TF-IDF baselines. Both models beat random performance, but RoBERTa does not uniformly outperform the lexical model: it improves social-direction prediction while collapsing toward the majority neutral class on economic direction. A smaller smoke experiment verifies the B5 and B6 comparison pipeline, but B6 underperforms B5 under constrained training. The results show that class imbalance and data access, more than architecture alone, are the central obstacles for reliable multidimensional ideology prediction.

1 Introduction

Ideology detection is the task of predicting political viewpoints from text. It matters because contemporary political discourse increasingly appears on social media platforms where researchers want to study public opinion, ideological fragmentation, political communication, and community reactions to major social issues. A useful system should do more than predict a single liberal or conservative label. Political text can express a stance toward a target, use a policy frame, mix economic and social views, or remain genuinely ambiguous.

This project asks whether weakly supervised transformer models can predict multidimensional ideology from social media text while remaining useful under topic, community, and platform shift. That question is motivated by Liu et al. (2023), who show that ideology should

be modeled through multiple facets rather than collapsed into a single axis. It is also motivated by Chen et al. (2023), who show that ideology classifiers can overfit topical content and biased supervision instead of learning more stable ideological cues.

The project began with a broader Reddit-centered design. The intended pipeline used MITweet for high-quality multifaceted labels, Reddit Politosphere for broad political community data, and Reddit Political Compass signals for weak economic and social ideology supervision. In practice, the strongest completed empirical result comes from MITweet because the Politosphere download was blocked in the available environment. The Reddit weak-supervision branch is therefore presented as implemented infrastructure and smoke-tested methodology, while the MITweet experiment is the main real-data result.

The main finding is cautionary. A pretrained transformer helps when the label distribution is reasonably balanced, as it does for social direction. It fails badly when class imbalance is extreme, as it does for economic direction, where the neutral class dominates. The smaller B5 and B6 smoke experiment also shows that a richer hierarchical model is not automatically better when data and training are limited. These results support a careful interpretation: before claiming architectural gains, the project needs class-weighted or focal losses, stronger per-head balancing, and a full real-data weak-pretraining run.

2 Related Work

Liu et al. (2023) introduce MITweet, a high-quality Twitter dataset annotated for ideological relevance and direction across multiple domains and facets. MITweet is central to this project because it provides direct support for multidimensional ideology prediction. Hofmann et al.

(2022) introduce the Reddit Politosphere, a large text and network resource covering political discussion communities over time. That dataset motivates the project focus on community structure and out-of-community generalization.

Colacrai et al. (2024) study Reddit ideology through economic and social axes, which aligns closely with the economic-direction and social-direction heads in this project. Rostami et al. (2025) provide PolitiSky24, a Bluesky stance dataset centered on the 2024 U.S. presidential election. Although PolitiSky24 is stance-focused rather than a full ideology dataset, it motivates future cross-platform evaluation beyond Reddit and Twitter-style text.

Liu et al. (2019) introduce RoBERTa as a strong pre-trained transformer encoder for text classification. However, political prediction is especially vulnerable to shortcut learning. Chen et al. (2023) argue that models trained from scarce and biased supervision should learn how ideology is expressed rather than merely memorize what topics appear in a dataset. Disagreement-aware NLP is also important for this project. Leonardelli et al. (2025) frame annotator disagreement as a meaningful learning signal, which fits political posts where ambiguity, mixed stance, and target framing can produce legitimate disagreement. Chang and Masterson (2020) show that lexical and sequence models can remain competitive for political text classification, which is why simpler baselines remain necessary.

3 Methodology

The project models ideology through eight prediction heads: relevance, target, stance, frame, economic direction, social direction, intensity, and ambiguity. Relevance identifies whether a post is political. Target, stance, and frame describe the discourse structure of the post. Economic direction and social direction capture the multidimensional ideology axes. Intensity captures how strong the ideological signal is, and ambiguity captures uncertainty or disagreement.

For the real-data branch, MITweet labels were projected into the heads that can be cleanly supported by that dataset: relevance, economic direction, and social direction. The local MITweet file contains 12,594 labeled posts. For the Reddit branch, the codebase implements ingestion, weak-label construction, gold annotation folding, counterfactual editing, model training, prediction export, calibration, paired bootstrap testing, and error analysis. The intended weak-supervision stage uses large Reddit re-

sources to learn broader political language patterns before gold fine-tuning.

B1 is a TF-IDF logistic-regression baseline. It uses balanced class weights in the real MITweet run, which becomes important for imbalanced ideology labels. B2 is a RoBERTa-base classifier fine-tuned on the available MITweet labels. B3 through B5 are flat transformer variants designed for weak pretraining and gold fine-tuning. B5 includes decomposition heads for target, stance, frame, intensity, and ambiguity. B6 is the hierarchical representation model. It maps text into a latent ideology representation and jointly predicts discourse and ideology heads.

The model supports both hard and soft labels. A hard label is a single target class, usually the majority annotator decision. A soft label is a probability distribution over labels, such as two annotators selecting support and one selecting mixed. Hard labels support ordinary classification training, while soft labels preserve disagreement. This matters for political text because some posts are legitimately mixed or ambiguous rather than simply mislabeled.

4 Experimental Design

The original experimental design evaluated whether the final model generalized across topics, communities, and platforms. The completed real-data experiment compares B1 and B2 on MITweet with three seeds. Evaluation uses macro-F1 because the ideology labels are imbalanced. The comparison also uses paired bootstrap testing over shared validation posts, which estimates whether one model outperforms another after resampling examples with replacement. Calibration is evaluated by comparing model confidence on correct and incorrect predictions.

Table 1 shows the real MITweet results. Both B1 and B2 beat random performance. B2 does slightly better on relevance and social direction, but B1 does substantially better on economic direction. The reason is label imbalance. Around 91% of economic-direction labels are neutral, and the RoBERTa model largely collapses toward that majority class. The balanced lexical baseline handles this imbalance better.

The paired bootstrap comparison gives the same interpretation. B1 significantly beats B2 on economic direction for all three seeds. On social direction, B2 beats B1 for one seed and ties for the other two. This means the transformer is useful when the label distribution gives it enough signal, but it is not robust to imbalance by default.

The confidence results in Table 2 show that correct B2 predictions are more confident than errors for every pro-

Label	B1 TF-IDF	B2 RoBERTa	Random
Relevance	0.708 ± 0.077	0.743 ± 0.064	~ 0.41
Economic	0.459 ± 0.021	0.341 ± 0.017	~ 0.21
Social	0.534 ± 0.015	0.579 ± 0.022	~ 0.34
Mean	0.567 ± 0.033	0.554 ± 0.024	0.314

Table 1: Three-seed MITweet results. Values are macro-F1 mean and standard deviation.

Label	Correct	Error	Gap
Relevance	0.957	0.849	+0.107
Economic	0.959	0.791	+0.168
Social	0.686	0.577	+0.109

Table 2: B2 confidence on correct and incorrect MITweet predictions.

jected MITweet label. This suggests that calibration and abstention are useful directions. The largest confidence gap occurs for economic direction, which is also where the model struggles most.

The B5 and B6 comparison was verified on a small synthetic smoke setup. This experiment is not a final real-data claim, but it verifies the prediction schema, supported-label macro-F1, paired bootstrap, and error analysis pipeline. Table 3 shows that B6 underperforms B5 overall under constrained training. Relevance is similar, but B6 is weaker on the ideology heads. The paired bootstrap comparison over 20 shared validation posts supports B5 over B6 for stance, intensity, and ambiguity, while the other labels are ties.

Figures 1 through 4 summarize the experiment artifacts. The heatmap shows model performance across evaluation regimes. The ablation chart shows how much individual components or diagnostics changed the measured result. The robustness panel shows noise sensitivity and counterfactual consistency. The error profile shows where failures concentrate and what content types dominate the sampled errors.

5 Limitations

The full Reddit weak-supervision experiment could not be completed because Politosphere access was blocked in the available environment. As a result, B3 through B6 should be treated as implemented and smoke-tested rather than fully evaluated on the intended Reddit scale.

The MITweet experiment is real-data evidence, but it only covers the labels that can be cleanly projected from MITweet: relevance, economic direction, and social

Metric	B5	B6
Macro-F1 mean	0.356	0.154
Relevance	0.467	0.473
Economic	0.322	0.194
Social	0.222	0.190

Table 3: Three-seed smoke-run results for the flat and hierarchical transformer variants.

direction. It does not fully evaluate target, stance, frame, intensity, or ambiguity in the same way as the Reddit annotation schema.

The visualizations summarize current artifacts, but some robustness and counterfactual consistency values are diagnostic estimates rather than direct model inference under every regime. The next experimental step is class-weighted or focal loss for the ideology heads, followed by a full real-data rerun once the Reddit weak-supervision data is reachable.

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Model Performance by Evaluation Regime

Macro-F1 by model and evaluation setting



Figure 1: Macro-F1 by model and evaluation regime.

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Ablation Deltas

Positive values indicate better performance or stronger support



Figure 2: Ablation deltas for model components and diagnostics.

Robustness and Counterfactual Consistency

Noise stress test and flat versus hierarchical consistency by edit type

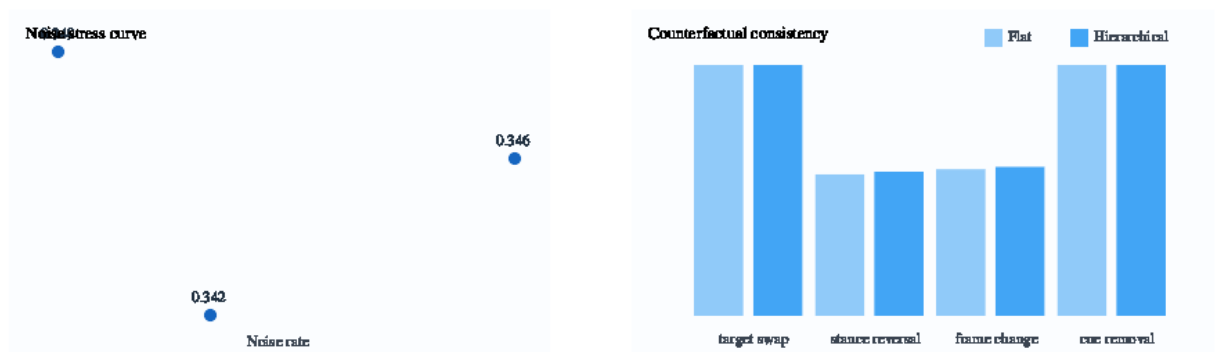


Figure 3: Robustness under noise and counterfactual edits.

Error Composition Profile

Stage-level failures and content taxonomy from 100 sampled error examples

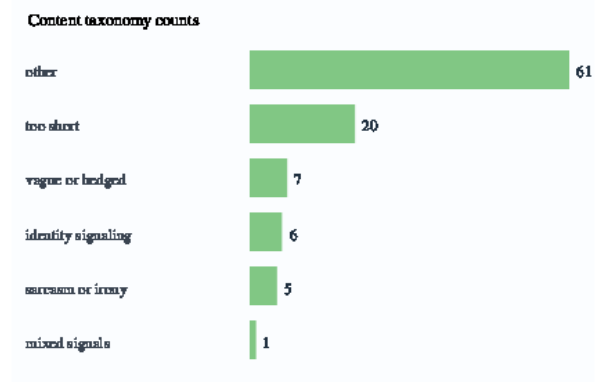
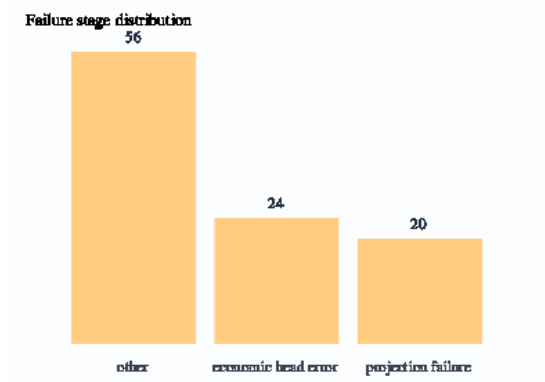


Figure 4: Error composition by failure type and content tag.